

WITHIN-FIELD VARIABILITY IN NUTRIENTS FOR SITE-SPECIFIC AGRICULTURAL MANAGEMENT IN IRRIGATED CORNFIELD



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HIGHLIGHTS

- Phosphorus and nitrogen in soil and plants varied during the growing season across the cornfield.
- Spatial variability in nutrients caused spatial variability in plant growth and crop yield across the cornfield.
- Hydrological characteristics and soil hydraulic properties were responsible for spatial variability in nutrients.
- Need to consider spatial variability in nutrients in delineating management zones before adopting management practices.

ABSTRACT. *The adoption of precision agricultural practices in Alabama has been increasing in recent years. Although challenging, understanding nutrient variability in agricultural fields is important for site-specific management. This study investigated phosphorus (P) and nitrogen (N) variability in an irrigated cornfield located in the Town Creek Watershed of Tennessee Valley Region of Alabama, USA during the 2019 growing season. Three different irrigation management zones (high yield-HY, moderate yield-MY, and low yield-LY) were delineated based on ten years of historical records of crop yield, soil texture, and measured topography using the Management Zone Analyst (MZA) software. The soil samples at 0 to 15 cm, 15 to 30 cm, and 30 to 60 cm depths were collected five different times during the 2019 growing season. Both manure and inorganic fertilizer were uniformly applied during the growing season. At all soil depths, HY and MY zones had higher soil nutrient concentrations than the field average nutrient concentration. However, the LY zone had below field average soil nutrient concentration in the entire soil profile. The plants (above-ground) in HY and MY zones had higher than average nutrient concentrations and the plants in the LY zone had lower than average nutrient concentrations throughout the cornfield. This site-specific nutrient variation in these zones was due to within-field terrain attributes, differences in soil properties, and surface runoff losses. A significant difference was recorded in the corn yield in the HY and LY zones of the field. Two extreme groups of nutrients were found in the cornfield similar to irrigation management zones (HY and LY). A zone with a higher yield also had higher soil and plant nutrients in the zone (HY), and a zone with a lower yield also had lower soil and plant nutrients in the zone (LY). Incorporating nutrient variability for site-specific management in management zone delineation can help with reducing nutrient application, nutrient loss, and improving yield. This study supports the adopting of precision agricultural management practices based on in-field nutrient variability.*

Keywords. *Corn, Crop yield variability, Irrigation, Management zones, Nitrogen, Nutrient variability, Phosphorus, Site-specific management, Topographical wetness index (TWI).*

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The world population is approximately 7.9 billion in 2022 with an average population growth rate of 0.9% annually (U.S. Census Bureau, 2022). Increasing population demands for increasing crop yield. Precision management of agricultural practices, especially nutrient and water management, during a growing season can help increase crop yield. Site-specific crop management is an important element of precision agriculture, which

aims to manage within-field variability, increase crop productivity, reduce average production costs per unit while optimizing resources, and foster environmental sustainability, thus improving land-use efficiency (Serrano et al., 2011). Farmers have been conventionally adopting a single rate for fertilizer or manure and irrigation application within a field. However, this uniform application leads to excessive nutrients or water in some areas and inadequate nutrients and water in others due to variability within the field (Mallarino and Witty, 2004). Although farmers have been adopting uniform applications for irrigation and nutrients, they have been facing the issue of crop yield variability across the field and inadequate productivity in some parts compared to others. Such agricultural systems are inefficient and need to be improved with the aim of precision agricultural management. Irrigated acres are increasing and the irrigated land has expanded by 5% in the last 15 years in the United States (USDA-NASS, 2019). However, the southeastern USA represents only 10% of the total irrigated land in the United States. Although Alabama has a relatively small amount of irrigated land, irrigation adoption has increased by 25% from 2012 to 2017 (USDA-NASS, 2019). The distribution of precipitation over the growing season is important to avoid water stress and meet crop water demand (Kumar et al., 2021b). Therefore, irrigation adoption has been increasing to help farmers improve water use efficiency, avoid water stress, maintain adequate soil moisture, and reduce nutrient loss. Thus, knowing the within-field variability in soil nutrients, adopting nutrient management practices (Serrano et al., 2011), and appropriate irrigation practices can help with improving nutrient availability to plants, reducing contamination of surface water due to surface runoff, and reducing pressure on fresh water availability during the growing season (Yost et al., 2019).

Soil nutrients are significant elements that support plant growth and crop productivity. Macronutrients (Phosphorus-P, Nitrogen-N, and Potassium-K) are required in large quantities and are the focus of many interventions, unlike micronutrients (Iron-Fe, Manganese-Mn, and Zinc-Zn, etc.) that are required in small quantities. The deficiency of micronutrients can result in severe crop failure, while excess levels can lead to health hazards (Mathew et al., 2016). Inadequate management of macronutrients limits crop yield and causes loss of soil productivity. Proper and effective management of macronutrients and assessment of their effects on crop yield and environmental quality requires an understanding of their variability in concentration across the field (Morales et al., 2014). Phosphorus (P) and nitrogen (N) are essential macronutrients for plant growth, which affect crop yield. However, excess application of these nutrients in agricultural fields leads to contamination of surface and groundwater (Hanrahan et al., 2019) locally, regionally, and globally. Several factors are responsible for nutrient loss from agricultural fields, including fertilizer application method, rate, and timing; irrigation rate; field topography; and soil properties (Hanrahan et al., 2019; Umali et al., 2012; Williams et al., 2018). To reduce nutrient loss in surface runoff, it is recommended not to irrigate soon after fertilizer application or apply fertilizer or manure in any form before an impending rain event (Schröder et al., 2004; Sharpley, 1997; Smith et al.,

2007). On the other hand, for surface application of manure, it is important to move these nutrients further into the soil where, for example, ammonia volatilization would be minimized. The longer the time between fertilizer or manure application and rainfall or irrigation, the smaller the nutrient loss in surface runoff (Schröder et al., 2004). Therefore, understanding nutrient variability is key for site-specific management to help minimize nutrient input, and discourage constant rate application throughout the field without considering variability in crop yield, which can benefit farmers by decreasing production costs with the added advantage of environmental protection.

The levels of P and N vary temporally and spatially within a field (Bogunovic et al., 2014; Gao et al., 2019; Su et al., 2018; Umali et al., 2012). Many studies have investigated spatial variation of nutrient levels in cropland, but few studies have focused on spatiotemporal variations within the same cropland during a growing season (Su et al., 2018). Therefore, more studies on variability in soil nutrients in cropland at various growth stages during a growing season are needed to help farmers to increase crop yield and reduce nutrient losses. Nutrient uptake for plant growth depends on nutrient availability in soil across the field (Gallardo and Paramá, 2007; Su et al., 2018). The accumulation of nutrient proportion in the plants depends on their availability in the soil (Razaq et al., 2017; Tewolde et al., 2019). If fertilizer application is not managed properly, the plant's ability to take other nutrients from soils is reduced, and the water quality is degraded due to surface runoff loss (Harmel et al., 2009). Also, inadequate P and N concentration in the plants negatively affects the plant and root growth development, as well as crop yield (Malhotra et al., 2018; Razaq et al., 2017). The leaching of nutrients in agricultural fields depends on the physical characteristics of soil (Chakraborty et al., 2020) and the strength of binding forces between nutrients and soil (King et al., 2015). Understanding site-specific management of soil nutrients in agricultural fields has been a trending topic as of late, but one that has proven challenging to study around the globe (Su et al., 2018) for those interested in adopting or researching precision management practices. Various experiments have been performed to study nutrient variability from plot to watershed scales to compare nutrient levels between plots, fields, and watersheds. However, studies on temporal variability with implications on crop yield variability are still scarce, and the results from controlled experiments have limited applicability to the actual field. It is important to understand the within-field variation of nutrients at different depths and at different times and stages during a growing season (Bogunovic et al., 2014; Su et al., 2018). The nutrient uptake in plants varies according to the growth stage of a crop during the growing season (Su et al., 2018). The tasseling and silking stages in corn, for example, are high nutrient demanding periods during a growing season. Knowing the nutrient level in the soil at different times can help to reduce yield loss or improve yield management. Therefore, it is important to investigate temporal variations in soil nutrients in crop fields during the growing season. Due to spatial variability in crop yield, it is important to optimize the agricultural systems to achieve growing food demands by reducing the yield losses or improving yield

management. Precision agriculture aims to improve site-specific yield management and identify the factors responsible for within-field variability that reduce the crop yield across the field (Jiang et al., 2021; McEntee et al., 2019). Therefore, studies need to investigate the significance of prevailing factors (field attributes, soil properties, and hydrological characteristics) influencing the nutrient variability, which is necessary for enhancing site-specific recommendations in farming practices, improving yield management, delineating management zones, and contributing to the refinement of existing decision support tools. Precision farming studies need to fill the gap of inefficient yield productivity in some areas within the field due to poorly or uniformly managed agricultural systems.

Previous studies have focused on nutrient budgeting (quantification of nutrients before and after the crop) and variation in different agricultural fields (Bundy and Sturgul, 2001; Hanrahan et al., 2019). However, less is known about the nutrient variation during the growing season (at various stages of the crops) within the same cropland (Su et al., 2018). Therefore, this study investigated within-field nutrient (P and N) variability at different soil depths and its impact on crop yield in a 120 ha cornfield located in Alabama, USA during the 2019 growing season. We quantified levels of soil P and N at various times during the growing season to understand spatiotemporal variability across the field. The first objective of this study was to determine within-field soil nutrient variability at different times and depths in a crop growing period. We hypothesized that soil P and N vary greatly within a field even after adopting the same best management practice (BMP) throughout the field. The second objective was to identify the factors significantly affecting nutrient and crop yield variability. We hypothesized that crop yield is significantly correlated with nutrient variability during a growing season.

MATERIALS AND METHODS

STUDY AREA DESCRIPTION

This research was conducted in Town Creek (34°43'7.15"N, 87°23'9.36"W) Watershed of Lawrence County in the Tennessee Valley Region (TVR) of North Alabama, USA (fig. 1). The study region is classified as subtropical humid with average annual precipitation of 140 cm (Southeast Regional Climate Center, (2020); Kumar et al., 2021b). The study was conducted in a 120 ha area of a cornfield during the 2019 growing season. The growing season usually occurs from April to October and the distribution of total monthly precipitation over the 2019 growing season is shown in figure 2. Dekalb® DKC 66-97, a corn hybrid was seeded with a seeding rate of 84,000 seeds per ha. The plant and row spacing in the field were 15 cm and 76 cm, respectively. Both manure and inorganic fertilizers were uniformly applied in the field during the growing season. Poultry litter was disseminated uniformly on the soil surface at a rate of 4.5 Mg ha⁻¹ using a mechanical spreader before the seeding and inorganic fertilizer was applied uniformly two times during the crop growing period (table 1). The application rate of N and P in the poultry litter was 105 kg ha⁻¹ and 116 kg ha⁻¹, respectively. The first inorganic fertilizer application was applied at the rate of 67 kg ha⁻¹ for nitrogen and 45 kg ha⁻¹ for phosphorus and potassium at the time of seeding (table 1). The side-dress application was conducted at the rate of 201 kg ha⁻¹ for nitrogen. The study area was equipped with a remotely controlled 625 m long Reinke center pivot irrigation system (Reinke Manufacturing Co., Inc., Deshler, NE, USA) with a mid-elevation spray application (MESA) sprinklers fitted with pressure regulators. With the uniform irrigation system, Kumar et al. (2021b) found soil moisture variability in the study field during the 2018 growing season. Therefore, the center pivot was retrofitted with variable-rate irrigation (VRI) technology with solenoid valves to control water application during the 2019 growing season. The

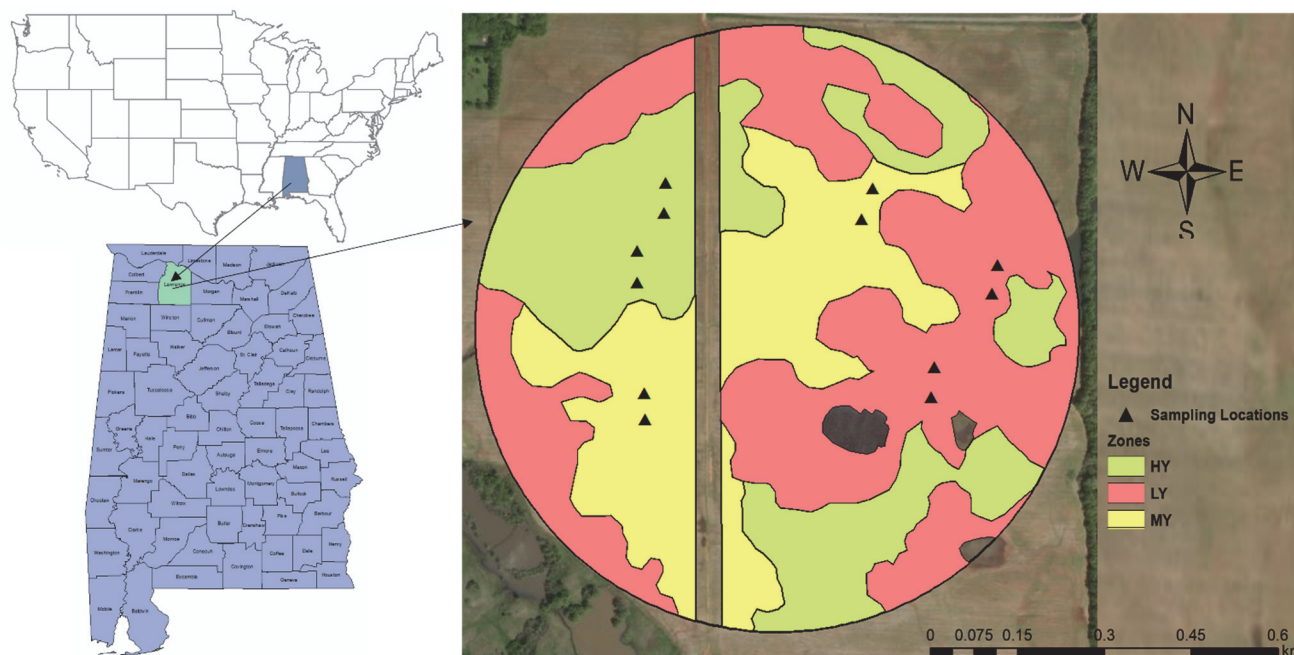


Figure 1. Study field with soil and plant sampling locations for nutrient analysis during the growing season.

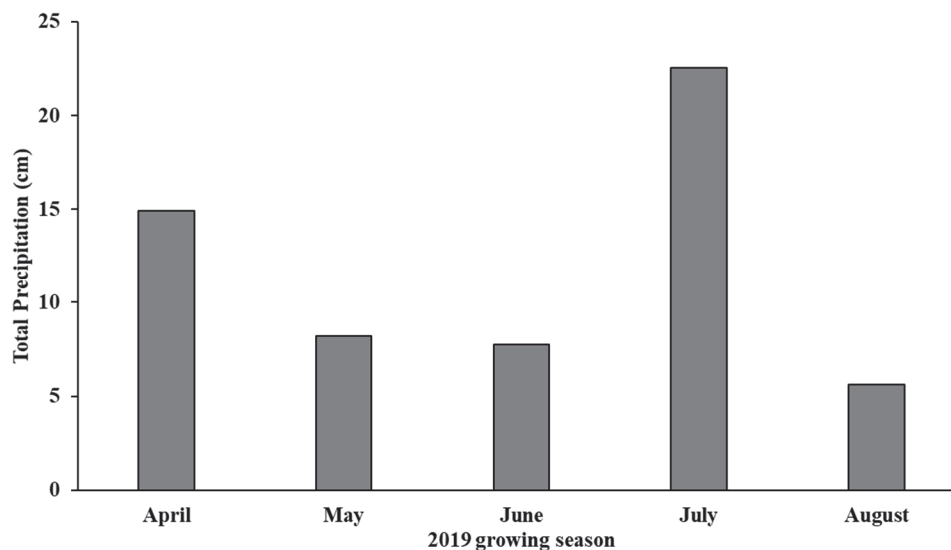


Figure 2. Total monthly precipitation received over 2019 growing season.

Table 1. Timeline of 2019 corn growing season in crop field.

SN	Task	Date	Remarks
1	1 st soil sampling	March 21, 2019	Pre-plant soil sampling
2	Poultry litter application	March 21, 2019	4.5 Mg ha ⁻¹
3	NPK application	March 28, 2019	67 kg ha ⁻¹ N, 45 kg ha ⁻¹ P ₂ O ₅ , and 45 kg ha ⁻¹ K ₂ O
4	Corn sowing	March 28, 2019	
5	2 nd soil sampling	April 28, 2019	Post-plant soil sampling
6	Side dressing	May 5, 2019	201 kg ha ⁻¹ N
7	3 rd soil sampling	June 22, 2019	Silking stage soil sampling
8	4 th soil sampling	August 1, 2019	Pre-harvest soil sampling
9	Corn harvesting	August 30, 2019	
10	5 th soil sampling	September 11, 2019	Post-harvest soil sampling

uniformity testing of the center pivot irrigation system was performed before the growing season with the catch can test (Dukes and Perry, 2006; Irmak et al., 2011; Ortiz et al., 2021). The field was under no-till management practices. Detailed and stepwise information for the Catch Can test is expanded upon in by Ortiz et al. (2021).

FIELD MEASUREMENTS AND DATA COLLECTION

Due to spatial and temporal variation in precipitation (Ridolfi et al., 2020), a weather station was installed near the field to measure precipitation locally during the growing season. Before soil sampling for nutrient analysis during the 2019 growing season, the field was delineated in different irrigation management zones based on topography, ten years of historical yield data, and soil texture (Kumar et al., 2021a) (fig. 1) using the Management Zone Analyst software (MZA, USDA-ARS, Columbia, MO, USA). A similar approach to delineating management zones was adopted by Jiménez et al. (2020) in South Alabama and Filho et al. (2020) in Central Alabama. This software uses unsupervised fuzzy classification to find homogenous areas with similar attributes using the input provided and creates boundaries at transitory areas between the zones (Fridgen et al., 2004). After processing the data through MZA software, three zones (namely, high yield-HY, moderate yield-MY, and low yield-LY) were identified. The areas covered by HY, MY, and LY zones were 34 ha, 30 ha, and 50 ha, respectively. To provide the information in MZA software, elevation data were collected using a StarFire™ 6000 real-time kinematic GPS

receiver (Deere and Company, Moline, IL, USA) mounted to the farmer's grain combine. The field varied in elevation from 168 m to 178 m above the mean sea level. The corn grain yield was collected using yield monitors mounted on a Combine with differentially corrected real-time kinematic (RTK) GPS systems. Based on historical yield data collected by the farmers, we found substantial yield variability across the field. Also, the corn grain yield for the 2019 growing season varied from 10125 kg ha⁻¹ to 18829 kg ha⁻¹ with an average of 14258 kg ha⁻¹ in the HY zone, from 9450 kg ha⁻¹ to 17317 kg ha⁻¹ with an average of 13941 kg ha⁻¹ in the MY zone, and from 3139 kg ha⁻¹ to 14301 kg ha⁻¹ with an average of 10575 kg ha⁻¹ in the LY zone. This corn grain yield was adjusted to 15.5% moisture (Tewolde et al., 2021). Soil samples from several locations at 0 to 15 cm, 15 to 30 cm, and 30 to 60 cm soil depths were collected in the field to determine soil texture using the hydrometer method (Bouyoucos, 1962). The soil properties were determined for each depth. Average soil properties for each zone are summarized in table 2.

Watermark sensors were installed to monitor soil matric potential at 0 to 15 cm, 15 to 30 cm, and 30 to 60 cm depths. The site-specific deficit irrigation was prescribed within the field during the growing season. Irrigation was triggered when there was a deficit of about 2.54 cm (as the upper limit) of irrigation depth in the root zone as compared to field capacity in any zone within the field. Each zone received a deficit irrigation amount indicated by the sensor data in each zone. The reason to use 2.54 cm as the upper limit was the

Table 2. Soil properties, sand (%), silt (%), clay (%), soil organic matter (SOM, %), bulk density (g cm^{-3}), and pH in each zone. Within each zone, mean values ($\pm$ standard deviation) of various soil properties followed by the identical letter are not significantly different at the 0.05 significance level.

Zone	Sand (%)	Silt (%)	Clay (%)	Soil texture	SOM (%)	Bulk density (g cm^{-3})	pH
HY	16 $\pm$ 9.06a	47.93 $\pm$ 10.65a	36.07 $\pm$ 15a	Silty clay loam	3.94 $\pm$ 0.40a	1.60 $\pm$ 08a	5.82 $\pm$ 0.60a
MY	22.31 $\pm$ 6.55a	40.67 $\pm$ 8.06a	37.02 $\pm$ 9.52a	Clay loam	4.18 $\pm$ 0.69a	1.60 $\pm$ 06a	5.13 $\pm$ 0.55b
LY	30.36 $\pm$ 7.86b	28.87 $\pm$ 9.27b	40.78 $\pm$ 8.78a	Clay	4.50 $\pm$ 0.78b	1.67 $\pm$ 08a	5.35 $\pm$ 0.68b

maximum capacity of the center pivot irrigation system for applying irrigation in the field. The variability in precipitation and irrigation amount received in each zone was captured in soil matric potential data, which were used to determine irrigation depth to schedule irrigation. The field capacity and permanent wilting point were assumed at 33 kPa and 1500 kPa, respectively, due to the fine texture category of the soils. A total of four irrigations were scheduled in the cropland during the growing season. Although the total water requirement for corn in Alabama is about 64 cm (Butler and Srivastava, 2007), the total amount of water received in HY, MY, and LY were 69.72 cm, 70.10 cm, and 68.78 cm, respectively. However, the distribution of precipitation is very important over the growing season and has an impact on irrigation frequency and crop yield. Therefore, if the precipitation is received during a high-water demand period, irrigation amount and frequency will be decreased. However, if the precipitation is received at the beginning or near the end of the growing season, it does not have much impact on irrigation. According to Lee (2019) (a guide to corn production in Georgia), tasseling and silking are the highest daily water demand periods for corn during a growing season. In our field, the high-water demand period (tasseling and silking) occurred from mid-May to mid-June during the 2019 growing season, and only 3 cm of precipitation was received during this interval of high-water demand. The maximum precipitation was received at the beginning (April) and towards the end (July) of the growing season. Therefore, irrigation was required to meet the crop water demands. All irrigation events happened between May and June in the cornfield. However, no significant differences were observed in the total water received by the HY, MY, and LY zones. Similar conditions have been observed by Filho et al. (2020) in Alabama and Irmak et al. (2012) in Nebraska, USA and they triggered irrigation during such period too.

After delineating the zones, soil samples were collected in the proximity of the Watermark sensors for nutrient analysis (fig. 1). A total of 540 (3 depths $\times$ 3 replicates $\times$ 3 zones $\times$ 4 locations $\times$ 5 sampling times) soil samples at four different locations in each zone with three replicates for each depth were collected for analysis before sowing to after harvesting the corn. The rationale behind the multiple locations was to capture maximum variability and reduce the uncertainty in soil nutrient data. Three replicates of collected soil samples were composited to homogenize for individual depth at each location. Plant samples (above-ground) were also collected at different corn stages in each management zone for nutrient analysis in the plants (Geisseler et al., 2017; Jones and Case, 1990). A grid of 1 m $\times$ 1 m was used in each zone for plant sampling during the growing season. According to the guidelines recommended by Geisseler et al. (2017) and Jones and Case (1990) for nutrient analysis in corn plant

samples, the whole plants in the grid at the V4 stage were collected as the first plant sampling. For the second and third plant samplings at the tasseling and silking stages, leaves opposite and below primary ears were collected. The plant heights and leaf area index (LAI) were also recorded in both zones at the silking stage of corn during the growing season. The LAI was measured using an LAI-2200 Plant Canopy Analyzer (LI-COR Environmental, Lincoln, NE, USA). Undisturbed soil samples were collected in each zone to develop soil water retention curves using an HYPROP measurement system (Peters et al., 2015). These undisturbed soil samples were collected using 5 cm deep and 8 cm in diameter stainless-steel rings. For a detailed procedure for developing soil water retention curves, readers can follow Lena et al. (2021, 2022).

SAMPLE PROCESSING

In the next step, the composited soil samples were oven-dried and ground using a soil grinder and passed through a 2-mm sieve for chemical analysis. The grinder was cleaned with a vacuum cleaner between the samples. The soil samples were analyzed for total P, total N, ortho-phosphorus (OP), and soil organic matter (SOM). The SOM was determined using the loss-on-ignition method (Wright et al., 2008) in the Soil and Water laboratory of the Department of Biosystems Engineering at Auburn University. For OP determination, colorimetric analysis (ammonium-molybdate-ascorbic method) was used (Ahmad et al., 2020; Harmel et al., 2009). The OP ions react with ammonium molybdate and antimony potassium tartrate under acidic conditions to form a molybdate-phosphate complex. The samples were extracted using distilled water (DI, pH = 6.95) for OP analysis with 3 g soil in a 50 mL centrifuge tube. The centrifuge tube was filled with DI water up to 30 ml (1:10 soil:solution). After centrifuging each sample at 4600 rpm for 30 minutes at 10 °C or less, samples were filtered through Whatman no. 42 filter paper in scintillation vials. The scintillation vials were stored in a refrigerator at 4 °C and analyzed within 24 hours for OP. Total P and total N in soil samples were determined by acid digestion using persulfate Kjeldahl digesting solution (Dayton et al., 2017, American Public Health Association, 1998). All of the standard solutions including blanks were also digested to keep a similar matrix for the analysis. A 0.5 g sample was digested in a 6 mL digesting solution and centrifuged before the analysis. A Lachat automated flow injection analyzer (Quikchem series 2) (Hach Company, Loveland, CO, USA) was used to analyze these nutrients in the soil samples. To ensure the accuracy of analyses and avoid discrepancies, we ran standard and blank solutions after every 15 samples. The plant samples were analyzed for total P using ICAP-Open vessel wet digestion Digi Block 3000 and for total N using LECO-

Nitrogen gas analyzer by Waters Agricultural Laboratories, Inc. (Camilla, GA). Plant samples were oven-dried at 80 °C. The dried samples were ground in a Wiley model 4 mill to pass through a 20 mesh (1.0 mm) screen before lab analyses. The mill was cleaned with a vacuum cleaner between the samples.

HYDROLOGICAL PROPERTIES ANALYSIS OF THE FIELD

To understand nutrient variability for site-specific management, the knowledge of within-field hydrology is important. Different terrain analyses using collected elevation data were conducted to evaluate the significant factors that can affect site-specific nutrient variability and that need to be considered in precision nutrient management. The slopes and topographical wetness index (TWI) (Beven and Kirkby, 1979) were determined using ArcGIS (version 10.3.1) (Esri, Redlands, CA, USA). The maps of TWI and slope were developed to understand the topographical variation in these three different zones (figs. 3a and 3b). TWI explains the potential high and low water accumulation areas within the landscape. A low TWI value represents a low water accumulation and high slope area, and a high TWI value represents a high water accumulation and flat area with a low slope (Qin et al., 2011). Maestrini and Basso (2018) found the areas that allow water accumulation without runoff and waterlogging exhibit large TWI values. The drainage lines (fig. 3b) were also delineated to understand the direction of the water flow.

STATISTICAL ANALYSIS

In this section, we will use the following indices for notation:

- The sampling time is indexed by j , with $j = 1, 2, \dots, N$ (where $N = 5$, number of sampling times);
- The sampling zone is indexed by k , with $k = 1, 2, 3$ and the specific sampling location is indexed by i , with $i = 1, 2, \dots, M$ (where $M = 12$, number of sampling locations);
- The sampling depth is indexed by l , with $l = 1, 2, 3$.

Considering these notations, it is assumed that the samples analyzed for nutrients are independent of each other at the same depth (for example, 0 to 15 cm depth soil samples

are independent between all locations). This assumption is valid because any two sampling locations were at least approximately 100 m apart. In addition, we used C as a common symbol for nutrient concentration in this study. The nutrient concentration C_{ijl} at each sampling location i , each day j , and each depth l was used to calculate different statistical features to understand the variability in nutrients during the growing season. The spatial mean ($\bar{C}_{jl}$) for each sampling time was calculated

$$\bar{C}_{jl} = \frac{\sum_{i=1}^M C_{ijl}}{M} \quad (1)$$

The temporal mean ($\bar{C}_{il}$) was calculated as

$$\bar{C}_{il} = \frac{\sum_{j=1}^N C_{ijl}}{N} \quad (2)$$

The spatial standard deviation ($\sigma(C_{jl})$) and coefficient of variation ($CV(C_{jl})$) of nutrients for each sampling time is given by

$$\sigma(C_{jl}) = \sqrt{\frac{1}{M-1} \sum_{i=1}^M (C_{ijl} - \bar{C}_{jl})^2} \quad (3)$$

$$CV(C_{jl}) = \frac{\sigma(C_{jl})}{\bar{C}_{jl}} \quad (4)$$

In the same way, the temporal standard deviation $\sigma(C_{il})$ and coefficient of variation $CV(C_{il})$ can be calculated for each location. The nutrient data were normalized based on the average concentration to understand the variation at different sampling locations and at different sampling times in the cropland during the growing season. The relative difference, $\delta(C_{ijl})$, for each location and time was defined as

$$\delta(C_{ijl}) = \frac{(C_{ijl} - \bar{C}_{jl})}{\bar{C}_{jl}} \quad (5)$$

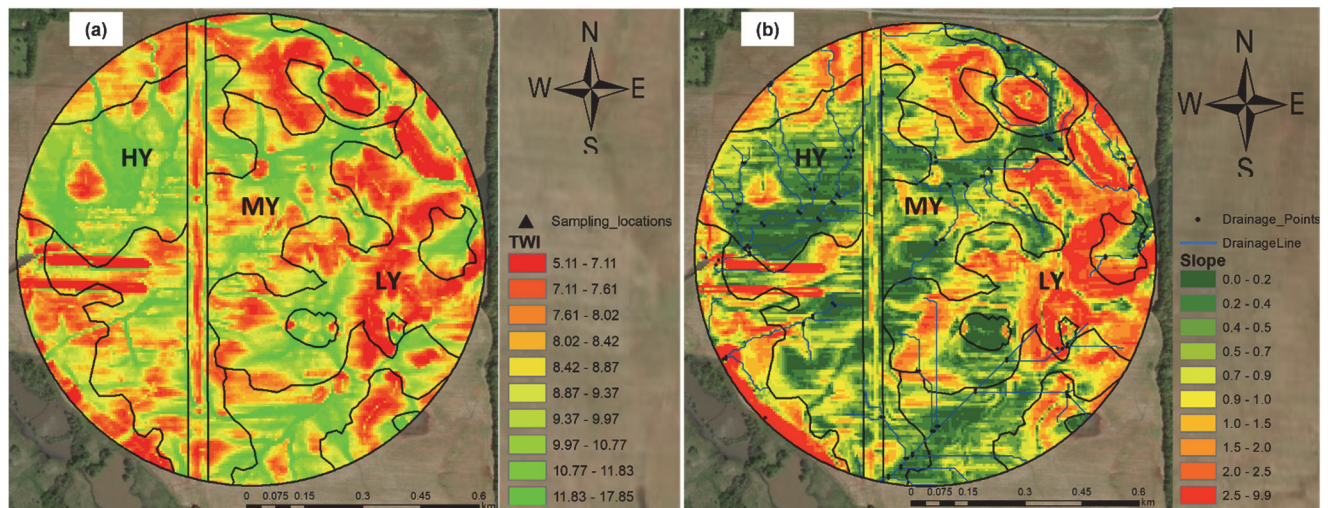


Figure 3. Terrain maps showing (a) Topographic wetness index (TWI) variation and (b) Slope (%) and drainage pattern in cornfield.

Based on the relative difference, the mean relative difference ($\bar{\delta}_{c,il}$) and standard deviation of relative difference ($\sigma(\bar{\delta}_{c,il})$) for each location (i) and each depth (l) can be given by

$$\bar{\delta}_{c,il} = \frac{\sum_{j=1}^N \delta(C_{ijl})}{N} \quad (6)$$

$$\sigma(\bar{\delta}_{c,il}) = \sqrt{\frac{1}{N-1} \sum_{j=1}^N (\delta(C_{ijl}) - \bar{\delta}_{c,il})^2} \quad (7)$$

To understand the zone-wise nutrient variation within the field during the growing season, the results were clustered according to zones based on sampling locations in the respective zones. The mean relative difference quantifies the departure of a nutrient in a zone from the average field nutrient and distinguishes whether the zone was lower or higher than the mean field nutrient during the growing period. A value of the mean relative difference in a zone at a certain depth greater or less than zero indicates adequate or inadequate conditions of nutrients concentration in the field, respectively. The analyses were carried out at each depth (l) independently. The Spearman's rank correlation (r_s) was computed to understand the association between the variables during the growing season.

Finally, in order to consider all effects jointly as well as the repeated observations over time, we employed a mixed linear model with nested random effects to assess nutrient variation between different zones and depths during the growing season. Since all effects are factors that can be represented by their respective indices, the model can be written as

$$C_{ijkl} = \beta_0 + \beta_j + \beta_k + \beta_l + \beta_{kl} + \gamma_k + \gamma_{k(i)} + \gamma_{k(i,l)} + \epsilon_{ijkl} \quad (8)$$

where β_m is the coefficient of the fixed effect for a general level m (m is general level for all β and γ coefficients such as 0, j , k , l), γ_m is the random effect for a general level m , and ϵ is the error term. More specifically, β_{kl} represents the fixed effect interaction between zone and depth while $\gamma_{k(i)}$ and $\gamma_{k(i,l)}$, respectively, represent the random effect of location (nested in the zone k) and depth (nested in the zone k and location i). Aside from taking into account the sampling time through the coefficient β_j , this mixed linear model aims to account for and isolate the dependence in the data due to the repeated measurements over time for each depth, location, and zone. By doing so, it is possible to obtain more precise and reliable testing for the fixed effects of interest which relate to the effect of time, zone, and depth on the nutrient concentration during the growing season. This approach has been adopted and promoted in other soil science studies dealing with data collected over different locations in space (Slaets et al., 2021). All the statistical analyses were performed using R software at a 0.05 significance level. In addition, we also conducted Moran's I test (Plant, 2018) on the residuals of the model to evaluate the spatial autocorrelation among them. This test led us to state that the observations

have no spatial autocorrelation among them. The p-values (0.26, 0.29, and 0.49, for total P, OP, and total N, respectively) obtained were too high as compared to 0.05 to reject the null hypothesis of no-autocorrelation for each nutrient. The Bonferroni correction was used for multiple comparisons (Plant, 2018). From this point forward in this article, the corresponding nutrient symbols will replace general notation C . For example, if we discuss soil total P, we will use TP in place of C to explain our results.

RESULTS AND DISCUSSION

SOIL P VARIABILITY DURING THE GROWING SEASON

The mean relative difference of P at three soil layers (0 to 15 cm, 15 to 30 cm, and 30 to 60 cm) of each zone, to understand the variation in the entire soil profile, were computed during the growing season. Based on the relative difference approach and mixed-linear model, the spatiotemporal patterns in total P between the zones across the cornfield were significantly different. According to $\bar{\delta}_{TP}$ in soil total P, HY and MY zones had a positive value, and the LY zone had a negative value at all three soil depths. For an average condition of soil total P in any zone, $\bar{\delta}_{TP}$ should be zero. Therefore, it explained the adequate condition of soil total P in the HY and MY zones and inadequate conditions of soil total P in the LY zone compared to the average soil total P within the field during the 2019 growing season (fig. 4). The $\bar{\delta}_{TP}$ was +0.17, +0.29, and +0.37 in the HY zone in 0 to 15 cm, 15 to 30 cm, and 30 to 60 cm soil profiles, respectively, during the growing season. However, in the LY zone, $\bar{\delta}_{TP}$ was -0.46 in 0 to 15 cm, -0.53 in 15 to 30 cm, and -0.38 in 30 to 60 cm soil profile. This pattern of total P variability indicated that the LY zone was low in soil total P relative to the average total P concentration in the field during the growing season. The mixed model explained a similar pattern of variability in total P during the growing season. The significant differences in total P were noticed between HY and LY, and MY and LY zones; however, there was no significant difference between HY and MY zones during the growing season. The effects of depth, time, zone, and interaction between zone and depth were significant for soil total P in the study. The variation in the field topography, soil physical and hydraulic properties, and field hydrology (Plach et al., 2019; Reid et al., 2019) resulted in the variable soil P levels across the cropland field, which are discussed in the following sections.

The total P data showed that the spatiotemporal mean of total P varied during the 2019 growing season, with the mean (CV) of 553 (0.13) mg kg⁻¹, 617 (0.13) mg kg⁻¹, and 257 (0.20) mg kg⁻¹ in the 0 to 15 cm soil profile of HY, MY, and LY zones, respectively. In 15 to 30 cm soil profile, soil total P was 510 (0.10) mg kg⁻¹ in HY, 489 (0.16) mg kg⁻¹ in MY, and 185 (0.14) mg kg⁻¹ in LY zones. In 30 to 60 cm soil profile, soil total P was 467 (0.10) mg kg⁻¹ in HY, 342 (0.27) mg kg⁻¹ in MY, and 208 (0.23) mg kg⁻¹ in LY zones. Based on descriptive analysis, substantial differences in within-field variability of soil total P were found across the zones during the growing season. The baseline (pre-plant

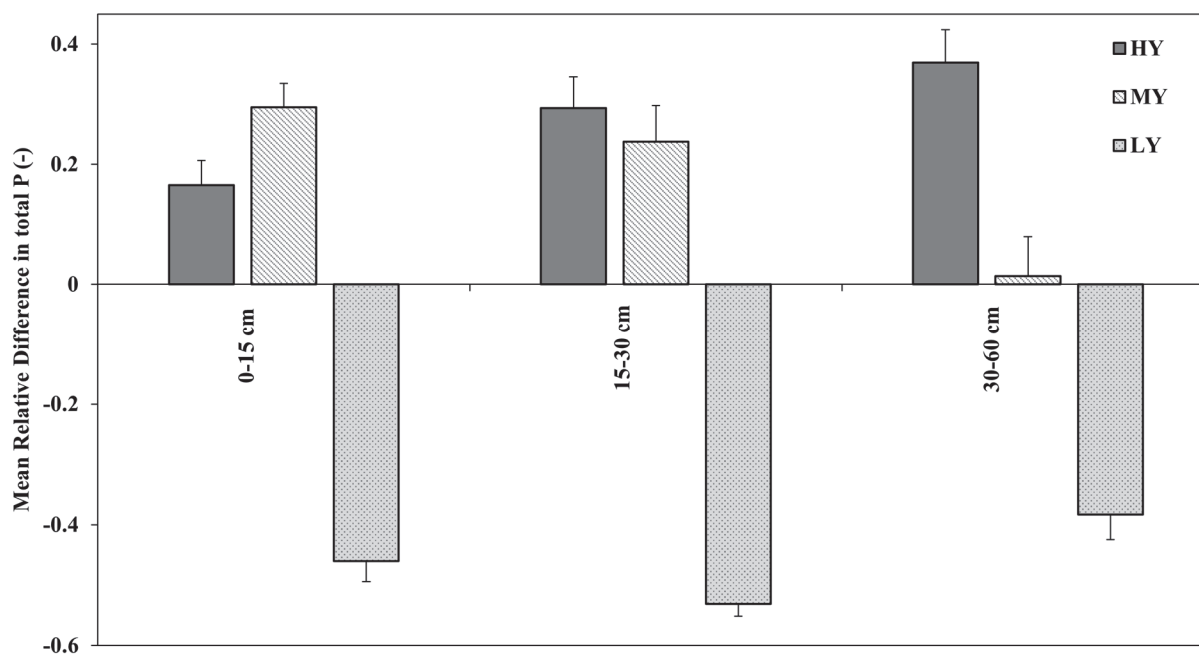


Figure 4. Mean relative difference in soil total P in zones at varying soil profiles during growing season. Error bars represent standard deviation in relative difference.

sample) total P concentrations in the 0 to 15 cm soil profile before the planting were 550 (0.14) mg kg⁻¹, 601 (0.13) mg kg⁻¹, and 234 (0.24) mg kg⁻¹ in the HY, MY, and LY zones, respectively. At the end of the growing season (post-harvest), in the 0 to 15 cm soil profile the total P concentration were 534 (0.12) mg kg⁻¹, 544 (0.16) mg kg⁻¹, and 236 (0.26) mg kg⁻¹ in the HY, MY, and LY zones, respectively. In the middle soil profile (15 to 30 cm), total P varied from 487 (0.12) mg kg⁻¹ at baseline to 530 (0.08) mg kg⁻¹ at end of the growing season in the HY zone. However, in the LY zone, the total P varied from 188 (0.13) mg kg⁻¹ at baseline to 180 (0.06) mg kg⁻¹ at end of the growing season. At the deepest soil profile (30 to 60 cm), total P varied from 488 (0.08) mg kg⁻¹ at baseline to 455 (0.16) mg kg⁻¹ at end of the growing season in the HY zone, from 331 (0.26) mg kg⁻¹ at baseline to 313 (0.29) mg kg⁻¹ at end of the growing season in the MY zone, and 212 (0.25) mg kg⁻¹ at baseline to 214 (0.25) mg kg⁻¹ at end of the growing season in the LY zone. The variability in soil total P showed that the HY zone had a greater and LY zone had a lower soil total P concentration than the average concentration of soil total P in all depths during the growing season, which means soil total P had different spatial variability between the HY and LY zones, and this needs to be considered in management zone delineation. It will help farmers to make decisions with precision farming practices in terms of variable nutrient management of zones within a field. It was inferred from the soil total P data that although applications (fertilizer and manure) were uniform in the field, total P was not uniformly available throughout the field during the growing season. Availability of P is rarely adequate for optimum growth and development of plants due to soil fixation (Malhotra et al., 2018). Soil OP also had similar spatial variability within HY and LY zones. Therefore, according to our analysis, lower soil total P than average in the LY zone might have restricted the release of

soil total P to OP and limited the supply to the plants in a similar proportion as the HY zone. Plants in the HY and LY zones did not accumulate nutrients in the same proportion due to differences in the availability of nutrients in soils, which affected the growth of the plants. Plants in the LY zone were smaller and had lower total P concentration compared to the HY zone due to lower P concentrations available in the LY zone, which impacted the crop yield (discussed in the section “impact of nutrients on crop yield”). Serrano et al. (2011) also found the similar pattern of variability in phosphate both greater and lower than average in a pasture field.

A similar trend of the mean relative difference in soil total P was also followed by soil OP (fig. 5). The HY zone had maximum, and the LY zone had minimum soil OP within the field. However, the MY zone was held between HY and LY zones. It explained the adequate condition of soil OP in the HY zone and inadequate conditions in the LY zone compared to average soil OP during the growing season (fig. 5). The mixed model also explained a similar pattern of variability in soil OP during the growing season. The HY zone was significantly higher in soil OP as compared to the LY zone. The deeper soil profiles were significantly lower in soil OP as compared to the topsoil profile (0 to 15 cm) in the HY zone. The effects of zone, depth, and time were significant for soil OP in the cropland. However, the depth did not have any interaction effect with the zones during the growing season. The mean relative difference showed an increasing pattern in soil total P and OP with soil depths in the HY zone. It showed the difference in soil P concentration increased with increasing soil depth as compared to average soil P concentration within the field corresponding to respective soil profiles during the growing season. However, the LY zone always had below average soil P concentrations during the growing season. Differences in physical and chemical soil

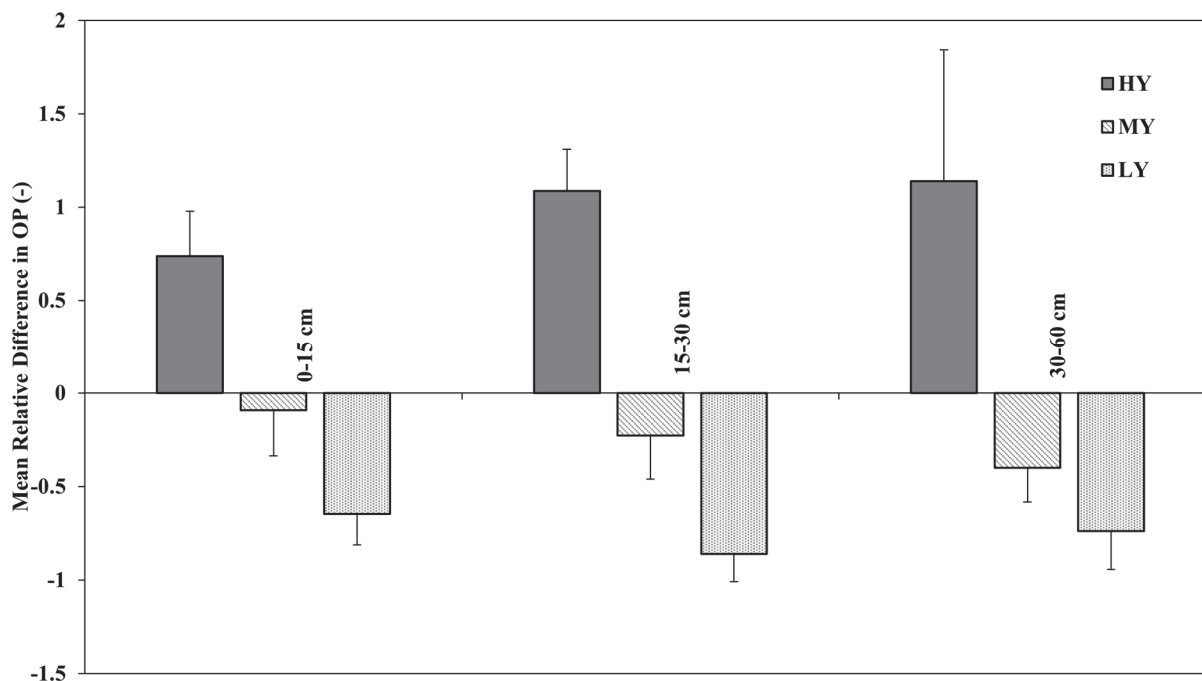


Figure 5. Mean relative difference in soil OP during the growing season. Error bars represent standard deviation in relative difference.

properties across the zones may have caused soil P variability during the growing season. The HY zone had a saturated hydraulic conductivity of 63 cm day^{-1} , and the LY zone had 9 cm day^{-1} . Within-field hydraulic conductivity variability in the HY and LY zones can help explain this variability. The downward movement of nutrients depends on water movement, for example, infiltration and percolation (Deftardarović et al., 2021). The slow rate of water movement in the LY zone decreased the flow of nutrients and increased opportunities for P fixation. Therefore, soil OP differences increase with increasing soil depth in the LY zone compared to the HY zone. Lower soil OP in the root zone reduced the nutrient uptake by plant roots, which affected the plant and root growth. The concentration of P affects plant reproduction, plant stalk strength, plant growth, and crop yield (Beegle and Durst, 2014). It was noticed that the plants in the LY zone were smaller in height than in the HY and MY zones. According to plant height measurements at the silking stage, the average plant height in the HY zone was 225 cm and 195 cm in the LY zone. Also, the average LAI was $4.13 \text{ cm}^2 \text{ cm}^{-2}$ and $3.34 \text{ cm}^2 \text{ cm}^{-2}$ in the HY and LY zones, respectively. P is required for the growth and development of plants (Dhillon et al., 2017). It is also necessary for energy transfer produced through photosynthesis to be used for reproduction (Malhotra et al., 2018). Availability of P in soil affects plant uptake and, consequently, plant and root growth (Razaq et al., 2017). This showed that an inadequate amount of P decreased the plant growth, and energy transfer for photosynthesis, and increased plant stress in the LY zone, which ultimately affected crop productivity. Similar results have been reported by Dhillon et al. (2017) and Malhotra et al. (2018). The applied fertilizers and manures were washed away due to surface runoff during the growing season since erosion patterns were observed in the LY zone.

SOIL N VARIABILITY DURING THE GROWING SEASON

The HY and MY zones had positive values of $\bar{\delta}_{TN}$ at all three depths (0 to 15 cm, 15 to 30 cm, and 30 to 60 cm) during the growing season. However, the LY zone had negative values for $\bar{\delta}_{TN}$ in the entire soil profile. The positive values describe the adequate condition of soil total N in the HY and MY zones and inadequate conditions of soil total N in the LY zone compared to the average soil total N within the field during the 2019 growing season, since the plant height and LAI were smaller in the LY zone as compared to the HY zone. In the HY zone, the $\bar{\delta}_{TN}$ was +0.01, +0.03, and +0.04, respectively, in 0 to 15 cm, 15 to 30 cm, and 30 to 60 cm soil profiles during the growing season. However, in the LY zone, $\bar{\delta}_{TN}$ was -0.04, -0.05, and -0.07, respectively, in 0 to 15 cm, 15 to 30 cm, and 30 to 60 cm soil profiles during the growing season.

A $\bar{\delta}_{TN}$ value close to zero represents the average soil total N concentration within the field during the growing season. The trend in total N variability showed that the soil total N was lower in the LY zone and greater in the HY zone compared to the average total N concentration within the field during the growing season. Tewolde et al. (2019) found that N accumulation in the plants in proportion highly depends on N availability in the soil. Our analysis showed that soil total N was not uniformly available throughout the field during the growing season. The HY zone had a higher proportion available as compared to the LY zone, which revealed that N also had the corresponding spatial variability in these zones similar to P. According to Tewolde et al. (2021), N becomes approximately 35% available after applying poultry litter in the same season and enhances the crop yield. In our study field, farmers uniformly applied poultry litter first and inorganic fertilizer afterward. Levels of N and crop yield were greater in the HY zone as compared

to the LY zone in all samplings. It was inferred from this analysis that the LY zone always had lower soil total N than the HY zone throughout the season, and variable management practices need to be adopted in such conditions for precision farming. Similar to P, N is also an important component for supporting plant and root growth (Razaq et al., 2017). Plant height and LAI at the silking stage were lower in the LY zone than in the HY zone, which revealed that N availability was lower in the LY zone for plant uptake and decreased plant growth and development as compared to the HY zone. The linear mixed model of soil total N also explained a similar trend across the field during the growing season. The linear-mixed model resulted in a lower soil total N in the LY zone than in the HY and MY zones during the growing season. It was also found that deep soil layers were significantly lower in soil total N than the topsoil profile across the crop field. However, no interaction between zones and soil profiles was reported in soil total N. The effects of the zone, depth, and time were significant for soil total N in the cropland. However, the depth did not have any interaction effect with the zones during the growing season. A significant effect of sampling time was observed in soil total N when the sampling was done during post-plant, silking, and pre-harvest times. Time as a fixed effect showed a significant effect on soil total N during the growing season. It was due to additional nitrogen application by farmers as they apply it conventionally during the growing season. The spatio-temporal mean of soil total N concentration in the soil profile of 0 to 15 cm was 3.75 (0.05) g kg⁻¹ in the HY zone, 3.81 (0.04) g kg⁻¹ in the MY zone, and 3.57 (0.05) g kg⁻¹ in the LY zone during the 2019 growing season. In the 15 to 30 cm soil profile, total N concentrations were 3.52 (0.05) g kg⁻¹ in HY zone, 3.50 (0.05) g kg⁻¹ in the MY zone, and 3.23 (0.06) g kg⁻¹ in LY zone. In the 30 to 60 cm soil profile, total N was 3.45 (0.05) g kg⁻¹ in the HY zone, 3.45 (0.05) g kg⁻¹ in the MY zone, and 3.10 (0.08) g kg⁻¹ in the LY zone. The positive increasing trend of the mean relative difference in

soil total N with increasing soil depths was observed in the HY zone. However, a negative increasing trend of the mean relative difference in soil total N with increasing soil depths was found in the LY zone during the growing season. This trend explained the high soil total N in the HY zone and low total N in the LY zone concerning the average soil total N within the field at the corresponding soil depths (fig. 6).

VARIABILITY IN NUTRIENTS RETAINED IN THE PLANTS DURING THE GROWING SEASON

Levels of soil nutrients provide major guidelines for fertilizer management in current and future growing seasons. However, the status of nutrients retained in the plants can be a complementary guideline to adopting BMPs for nutrient management and for comprehending the yield. We investigated the plant total N and total P for variability in nutrients retained in the plants at different stages during the 2019 growing season. According to the mean relative difference in plant total N and total P, the HY and MY zones had a positive value, and the LY zone had a negative value (fig. 7). It was found that nutrients accumulated in the plants had a greater concentration in the HY zone and a lower concentration in the LY zone compared to the average concentration of plant nutrients within the field. N and P are critical nutrients for plant growth and root development, which are important components to evaluate the effect of nutrients on both plant and root growth. Razaq et al. (2017) found that higher N and P concentrations in the plants and fertilizer application in the field significantly affected the plant height and root growth. Similar conditions were found in our study; plant growth and nutrients were lower in the LY zone and greater in the HY zone due to lower soil nutrient concentrations available within the LY zone than HY zone. The plant height was observed as 225 cm in the HY zone and 195 cm in the LY zone and 3.34 cm² cm⁻² in the LY zone at silking stage. The plants in the LY zone had the lower possibility of

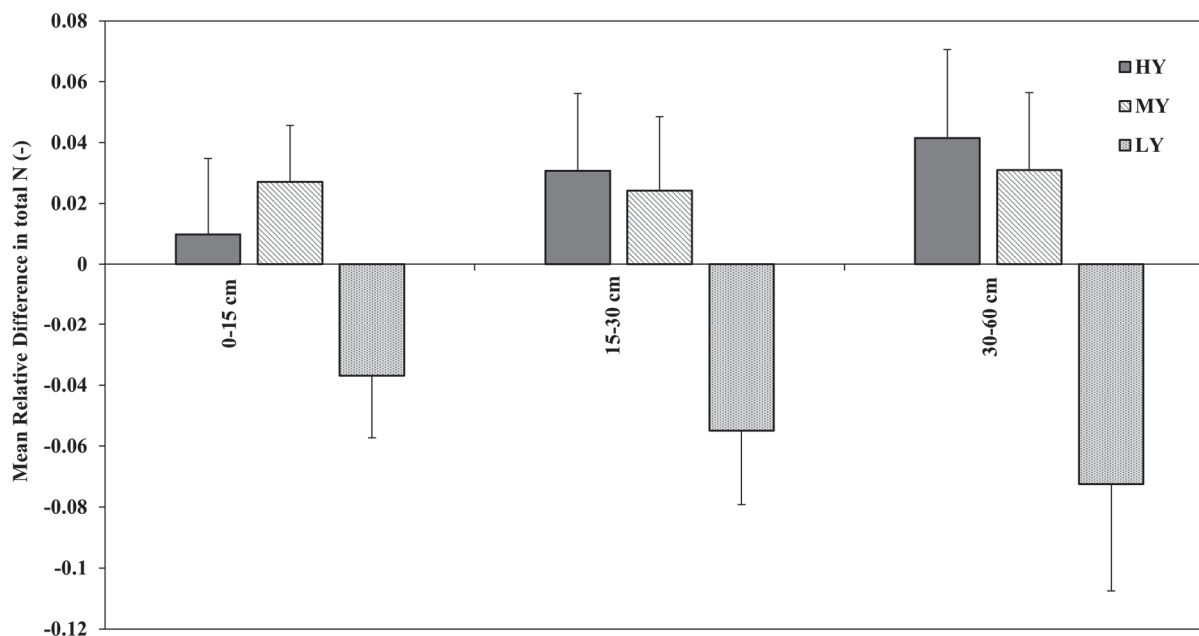


Figure 6. Mean relative difference in soil total N during the growing season. Error bars represent standard deviation in relative difference.

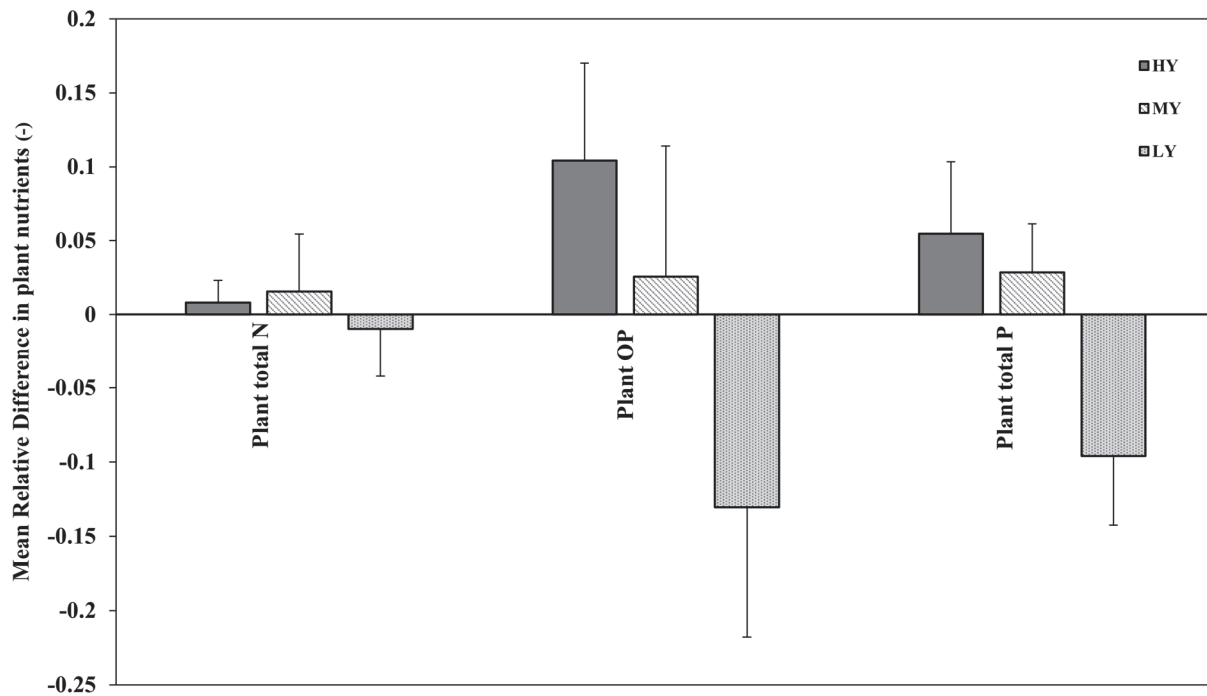


Figure 7. Mean relative difference in plant nutrients during the growing season. Error bars represent standard deviation in relative difference.

nutrient uptake due to lower nutrient concentration available in the soil as found in previous studies (Pereira et al., 2020; Razaq et al., 2017; Tewolde et al., 2021), which affected the plant height, growth, and productivity in the LY zone. Therefore, plants in the HY zone had adequate nutrients and the LY zone had inadequate nutrients during the growing season (fig. 7). The plant nutrients had a similar pattern of variability as in the soil nutrients obtained across the field. This revealed the negative impact of nutrient loss in the LY zone on plant growth and crop yield (fig. 8). Therefore, the results of this study showed that yield in the HY zone was related to adequate soil and plant nutrients in the HY zone, and yield

in the LY zone was related to inadequate soil and plant nutrients in the LY zone during the growing season.

FACTORS RESPONSIBLE FOR SITE-SPECIFIC NUTRIENT VARIABILITY

The above-mentioned nutrient variability in different management zones led to the investigation of different factors (e.g., topography, soil properties, and hydrology of the field) that caused variation within the field. The TWI map (fig. 3a) showed lower values of TWI in the LY zone which led to a lower concentration of OP (slow conversion of total P into OP due to lower wetness for the mineralization

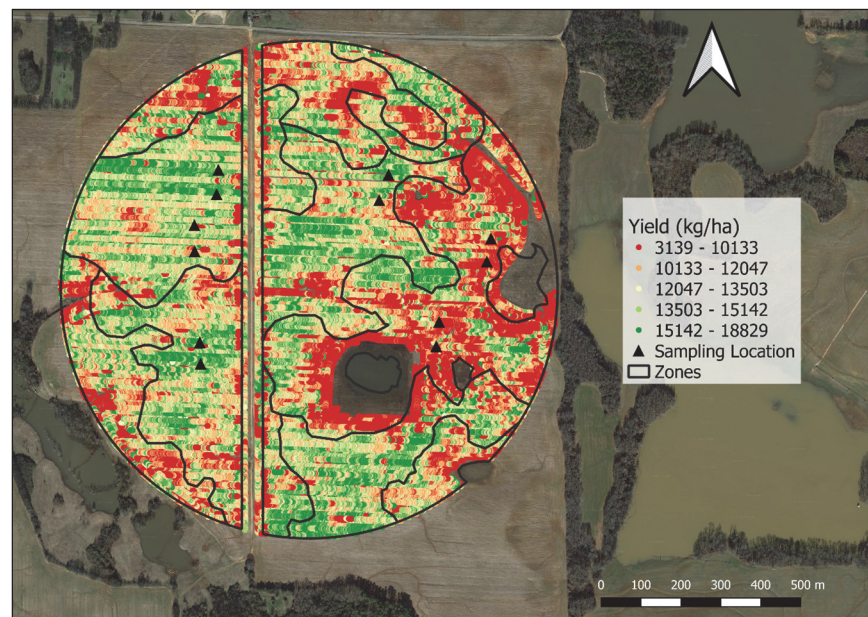


Figure 8. Crop yield variation in cornfield during 2019 growing season. Corn grain yield was adjusted to 15.5% moisture. Gray areas indicate areas of relatively low elevation in field.

process) during the growing season. Also, lower TWI in the LY zone led to lower total P and total N concentration in LY and higher TWI in HY led to higher total P and total N in the zone. This lower TWI affected the mineralization process that is affected by biotic and abiotic factors, including organic P substrate availability and the geochemical properties of soil (Shaw and Cleveland, 2020). Mineralization is a slow process, and this could have been an issue for a decade in the field, which resulted in low plant-available P and N in the LY zone. Ultimately, these long-term processes reduced the productivity in this part of the field. We also calculated r_s between TWI and the temporal average of soil nutrients to understand the relationship with TWI across the crop field (table 3). The correlation coefficients were significant in the soil profiles for soil nutrients during the growing season. Based on a positive correlation between TWI and nutrient concentrations, a higher TWI value in the HY zone was correlated with the higher nutrient concentrations in HY, and a lower value of TWI in LY was correlated with the lower nutrient concentrations in the LY zone. The soil P had a stronger correlation in upper soil layers than the deep soil layer, which explained the strong soil P dependency on TWI at the upper soil layer than in the deep soil layers. However, soil total N had a stronger correlation with TWI in the deep soil layers than in the upper soil layers, which implied the strong soil N dependency on TWI in deep soil layers.

Erosion gullies were observed in the LY zone during the growth stages and after harvesting, which revealed nutrient loss due to surface runoff and topographical variations in the field. The topographical variations can be understood using the slope variations in the field (fig. 3b). A high slope area increases the chance of lateral water movement and reduces infiltration (Huat et al., 2006). In our study field, the LY zone had a higher elevation and slope than the HY zone. Previous studies have found that higher elevation and steep fields tend to have higher erosion and runoff, lower infiltration, and lower soil productivity (Jiang and Thelen, 2004). We found similar results in our study that slope was greater and hydraulic conductivity was lower in the LY zone as compared to the HY zone. The lower hydraulic conductivity and greater slope contributed towards the generation of the surface runoff in the LY zone. The runoff generation was one of the dominant factors for lower nutrient concentration in the LY zone; also reported by Plach et al. (2018) and Smith et al. (2007) in their studies. Greater slope and lower TWI values in the LY zone did not create the conditions for water accumulation, which resulted in the loss of nutrients with generated surface runoff. However, the HY zone with greater TWI and lower slope had the greater water accumulation condition, which allowed the nutrients to be retained in the zone and reduced the loss due to no surface runoff. Therefore, the HY zone behaved as a pool for nutrients due

to no surface runoff and high-water accumulation, and no erosion patterns were observed during and after the growing season.

We also computed r_s between slope and the temporal mean of soil nutrient concentrations to explain the relationship of soil nutrients with slope across the cropland. The r_s were significant for all nutrients except soil OP at 30 to 60 cm soil layer (table 3). A negative correlation between nutrient concentrations and slope explained that a lower slope in the HY zone had higher nutrient concentrations in the HY zone, and a higher slope in the LY zone had lower nutrient concentrations in the LY zone. Similar relationships were obtained in earlier studies conducted in Michigan, USA (Jiang and Thelen, 2004). Less availability of soil nutrients in certain areas of the field reduced the crop yield. The weak binding of nutrients with soils due to dominant hydrological characteristics (TWI, slope, runoff) and soil physical properties in the LY zone enhanced losses. The surface runoff, TWI, erodibility, and slope across the field induced nutrient loss during the growing season. The hydrologic perspective also suggests that soil profile with higher hydraulic conductivity will have a high infiltration rate and reduced runoff and vice versa (Cai and Ugai, 2004; Duong et al., 2019; Qi and Vanapalli, 2015). According to soil hydraulic properties in all three zones, the LY zone had lower hydraulic conductivity, higher slope, and lower TWI. Based on water input data, the entire field received the same precipitation events and the LY zone received the least irrigation amount during the growing season; however, no significant differences were found in the irrigation amount at the end of the growing season. When the rainfall intensity is greater than the saturated hydraulic conductivity of the soil, surface runoff occurs along the slope (Huat et al., 2006). Saturated hydraulic conductivity for the LY zone (9 cm day^{-1}) induced surface runoff, thereby, causing nutrient loss during the growing season. Djodjic et al. (2004) also found that higher water movement down and laterally through the soil profile leads to leaching faster, compared to slow water movement, which can also be seen from the drainage pattern within irrigation zones of the field (fig. 3b). We also computed r_s between total P and OP in each of the zones during the growing season. The correlation coefficients were significant with 0.52 ($p < 0.0001$) in the HY zone and 0.42 ($p < 0.0001$) in the LY zone. We noticed from the correlation that HY had a better association between soil total P and soil OP than the LY zone during the growing season. Low water accumulation, an outward drainage pattern, surface runoff, and lower hydraulic conductivity limited water movement in the LY zone as compared to the HY zone, which had greater hydraulic conductivity, inward drainage pattern, and no runoff. Leaching of nutrients within the soil layers is driven by water infiltration. However, the field hydrological characteristics were not conducive to infiltration in the LY zone, which resulted in surface runoff with eroded nutrients from the LY zone. Lack of nutrients within the root zone reduced nutrient uptake, which negatively affected the plant height and growth during the growing season. The inward drainage pattern did not allow water to flow out as overland flow and retained the nutrients in the HY zone and worked as a nutrient pool. Therefore, lower slope and higher hydraulic conductivity in the HY zone

Table 3. Correlation coefficients between terrain attributes and soil nutrients in the cornfield. Asterisks (*) indicate significance at 0.05 level.

	TWI			Slope		
	0-15 cm	15-30 cm	30-60 cm	0-15 cm	15-30 cm	30-60 cm
Soil TP	0.74*	0.82*	0.76*	-0.84*	-0.78*	-0.56*
Soil OP	0.80*	0.76*	0.34	-0.71*	-0.52*	-0.18
Soil TN	0.64*	0.78*	0.81*	-0.68*	-0.78*	-0.80*

resulted in greater infiltration and nutrient leaching (fig. 3b) as compared to the LY zone. This increased the nutrient uptake in the root zone and the plants in the HY zone had greater nutrient concentration, which enhanced the plant growth in the HY zone than in the LY zone. The effect of nutrient uptake in plants was observed on plant height and LAI during the growing season. Plants in the LY zone were not developed in a similar proportion as in the HY zone, which ultimately caused the variability in nutrients level and affected the crop yield (this will be discussed in following section).

IMPACT OF NUTRIENT VARIABILITY ON CROP YIELD

The soil nutrients varied across the zones. Also, there was spatial variation in crop yield at the end of the season. The yield produced within the field during the 2019 growing season varied from 3,139 kg ha⁻¹ to 18,829 kg ha⁻¹. A significant difference ($p < 0.001$) was recorded in the crop yield in HY and LY zones across the field. The average yields (standard deviation) were 14258 (872) kg ha⁻¹ and 10575 (2022) kg ha⁻¹ in the HY and LY zones, respectively. The least yield was produced in the LY zone (red) (fig. 8). However, a higher yield was produced in the other two zones (HY and MY) of the field (green) (fig. 8). Adequate availability of nutrients in the soil increases the nutrient uptake by plants, and adequate nutrient concentration in the plants helps with plant height, growth, and productivity (Malhotra et al., 2018; Pereira et al., 2020; Tewolde et al., 2019). In our study, we found that the plant heights were 225 cm and 195 cm with corresponding LAI of 4.13 cm² cm⁻² and 3.34 cm² cm⁻² in the HY and LY zones at the silking stage of corn, respectively. Lack of nutrient availability in the LY zone limited the plant uptake and reduced the crop growth in the LY zone, which caused the corn yield differences across the field.

Plant Nutrients and Crop Yield

We also calculated r_s between yield and plant total P and plant total N during the growing season. The tasseling and silking stages of corn occur in the high nutrient demand period in a growing season. The plant total P at silking stage had a positive significant correlation of +0.63 ($p < 0.0001$), which indicated that increasing P increase crop yield. However, the correlation was not significant at the V4 and tasseling stages. The plant total N had a positive significant correlation of +0.85 ($p < 0.0001$) at the tasseling stage, however, a negative correlation of -0.63 ($p < 0.0001$) was at the V4 stage of the corn. It explained that yield had a positive correlation with the nutrient uptake by the plants during the growing season. The r_s between yield and plant nutrients indicated that higher nutrients retained in the plants at the tasseling and silking stages can increase corn yield.

Topographical Attributes and Crop Yield

We also investigated TWI and slope as responsible factors for crop yield variability in the field other than P and N. An area with higher TWI values resulted in a higher crop yield (Maestrini and Basso, 2018). We estimated +0.66 ($p < 0.02$), a strong r_s between the TWI and crop yield of the field, which had a dependency of crop yield on the topographical wetness of the field. Lower TWI led to low crop

yield, and in contrast, a higher TWI led to higher crop yield in the field. Slope and crop yield had a negative correlation of -0.85 ($p < 0.0005$), which explained that a higher slope in the field led to a lower crop yield of corn. Similar results have been found in a study conducted by Jiang and Thelen (2004) in a cornfield. Based on the above discussion, the LY zone had a greater slope and lower yield than the HY zone. A lower elevation and slope have greater water accumulation and greater yield productivity (Jiang and Thelen, 2004; Serrano et al., 2011). In our study field, TWI was also lower along with a greater slope in the LY zone and did not have enough water accumulation conditions to release soil nutrients for plant uptake unlike in the HY zone with a lower slope and greater TWI, which ultimately had an impact on plant height, LAI, and crop yield. Plants in the HY zone had greater nutrient concentrations, and plants in the LY zone can have the same potential of nutrients if the LY zone can be practiced in a different way to increase the nutrient concentrations in the soil and reduce losses during the growing season.

SUMMARY AND CONCLUSIONS

It can be concluded from the study that the soil nutrient availability and nutrient retained in the plants did not occur uniformly across the crop field, although fertilizers and manure were uniformly applied during the 2019 growing season. The crop growth data showed that plants grew proportionally to the soil nutrient availability throughout the field. The soil nutrients (P and N) had within-field variability across the cornfield during the growing season. Nutrients retained in the plant had a similar pattern of within-field variability in soil nutrients during the growing season. Nutrient variability was correlated with crop yield variability across the cornfield. The HY and MY zones had higher nutrients in soil and plants compared to the LY zone. The zone with greater nutrient concentrations compared to field average nutrient concentrations had a higher crop yield, and the zone with a lower nutrient concentration compared to the field average nutrient concentrations had a lower crop yield. Consistent management practices (irrigation, fertilizer, and manure) were applied across the field; however, different topographic attributes affected nutrient availability, loss, nutrient uptake, crop growth and development, and crop yield across the field. The TWI, slope, and soil hydraulic properties were found as factors responsible for within-field nutrient and crop yield variability across the crop field. Considering nutrient variability in management zone delineation can be an effective way to improve precision agriculture adoption during the growing season. Mainly, two extreme groups of nutrient variability (HY zone and LY zone) in soil and plants were identified, which can play an important role in the site-specific management of nutrients to improve crop yield. Therefore, it is of utmost importance to study nutrient variability across crop fields to increase crop yield, reduce loss of nutrients, and reduce negative environmental aspects. Improving and optimizing the lower crop yield areas similar to high yield areas can increase productivity from the same crop field during the same growing season.

IMPLICATIONS FOR SITE-SPECIFIC MANAGEMENT

Certain areas of a field may be more susceptible to nutrient loss (surface runoff loss in our study), lower nutrient availability for plant uptake, and, thus, reduce crop yield. For large cropland areas, executing uniform BMPs throughout the field fails to obtain uniform nutrient variability and yield across the crop field during the growing season. Implementation of different BMPs can improve nutrient variability and availability and can also help improve crop yield. Farmers need to use different fertilizer rates and strategies within zones of the field during the growing season. As a recommendation, to minimize nutrient losses, improve crop and root growth, reduce negative environmental aspects, and increase crop productivity, farmers can use one or multiple conservation practices described below.

- Apply nutrients to meet the agronomic requirements of the crop fields. The surface application of nutrient sources needs to be in the zones with a low potential for surface runoff.
- Sub-surface application of fertilizers or manures can help nutrients to be held in the soil for a longer period during the growing season, especially in the runoff susceptible areas. It can include subsurface band application of poultry litter and knifing-in the side-dress application of N.
- The use of multiple irrigation events with lower water application rates might help reduce nutrient losses in the zones which have high nutrient loss potential caused by surface runoff during a growing season. This can promote plant health and the field may have the same crop maturity time, so crop productivity may be increased.
- Drip irrigation can be a good irrigation practice in areas that have high nutrient loss potential through surface runoff.
- Fertigation can also be a good option for farmers to reduce energy consumption and increase nutrient availability and nutrient use efficiency.
- Cover crops in the lower nutrient and higher runoff generating areas may help enhance the soil water-holding capacity, increase nutrient adsorption in the soil, and improve field hydrologic and hydraulic properties.
- Conservation tillage with cover crops in the LY areas can reduce soil erosion, loss of sediment-bound nutrients, conserve soil water, build the soil structure, and improve water infiltration. However, tillage with no cover crops increases the disruption in soil structure, accelerating the surface runoff and soil erosion, which increase nutrient loss and crop yield loss.
- No-tillage with cover crops in the lower nutrient and higher runoff generating areas can help enhance the soil water-holding capacity, increase nutrient adsorption in the soil, and improve field hydrologic and hydraulic properties (such as water retention, infiltration, and hydraulic conductivity). In Alabama, cover crop residue left in the field reduced surface runoff

and increased water infiltration by 50% to 800% (SARE, 2007).

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